see in the headset's micro-display. According to BMW, vehicle repairs are now being completed 75% faster. And it also serves the purpose of on-the-job training of local technicians!

In a recent podcast, McKinsey senior expert Richard Ward explained, "BMW's new all-electric vehicle production line ran as a simulation for six months, building virtual cars on a one-to-one scale in the metaverse, before they actually did the final layout for the factory. And in the process of those six months, they changed the design about 30% from the original. They have not said publicly how much more efficient it was, but they did say that about 30% of what they thought was the world's best factory on day one of the simulation had to change in the process. These are people who build new factories daily, and they still found that level of learning from the simulation. So, why not practice first in the metaverse, where it's low-cost, you can do things in infinite ways, and you can make the impossible happen? Those types of things really offer people more efficiency and productivity."

Military training: Can't afford to learn on the job!

Governments across the world are experimenting with how the metaverse can be used for military training and strategising.

One of the oft-spoken-about examples is that of flight simulators used to train pilots. And now, real-time information overlay using AR headsets is being used to enable soldiers and technicians to explore weaponry, machinery, automobiles, and other military equipment. This enables them to learn at their pace. The faster learner gets to take the tests earlier! Strategies can also be formulated better using interactive virtual platforms rather than presentations or charts.

We see start-ups coming up with innovative solutions for the defence sector—and the Indian military forces are quick to recognise, collaborate, and utilise these.

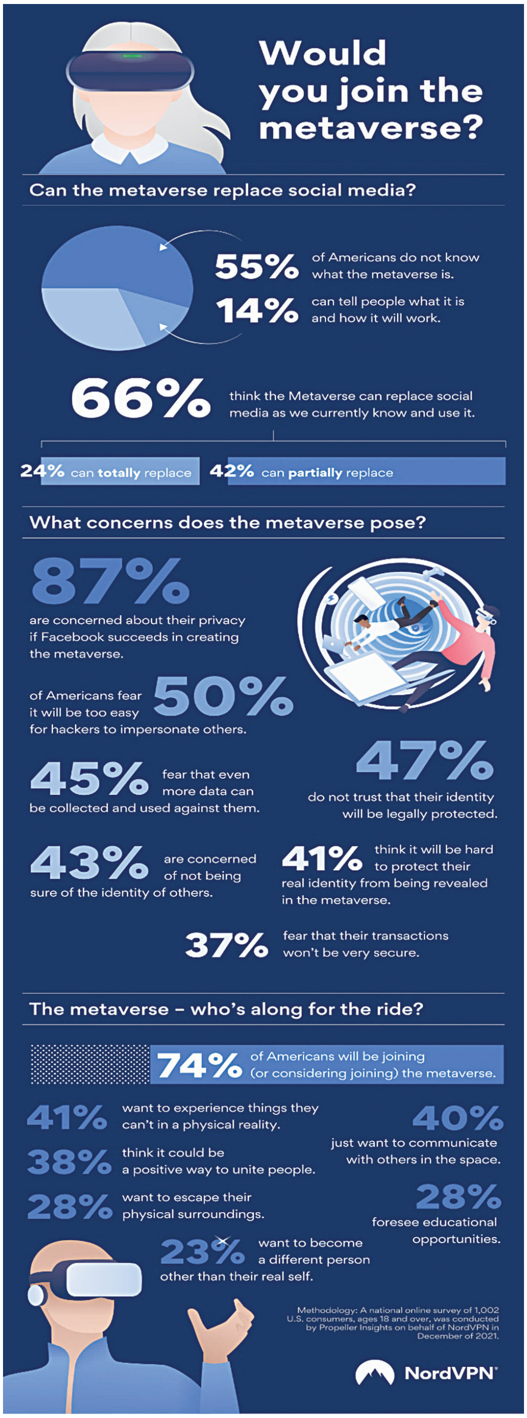

Figure 4: A survey conducted by Propeller Insights for NordVPN in the US (Courtesy: NordVPN)

One such homogenous solution is the see-through armour developed by AjnaLens. The AjnaESAS (Enhanced Situational Awareness System) helps overcome a key problem faced by the crew inside a tank. It is very difficult to navigate a tank using the small periscopic view. AjnaESAS uses an AR based head-mounted display and a 360-degree camera to offer the crew a 360-degree horizontal field of view. The camera system also has night vision and 4X zoom capabilities. AjnaESAS uses AI to help navigate the tanks through challenging terrains by eliminating blind spots and providing clear images even when the view is obscured by dust or smoke.

Another of their solutions, AjnaBolt, improves weapon systems with features like thermal sensors for night vision, friend or foe detection (popularly known as IFF—identification, friend or foe), light detection and ranging (lidar), blue force tracking (a GPS-enabled system that gives a real-time picture of the battlefield, which conventional maps cannot give), and so on. Fair-weather weapons can be effectively upgraded to all-weather weapons, providing visual assistance in locating, locking, and engaging targets. When it is combined with a holographic sandboxing system, it helps render a detailed 3D holographic map for planning and preparation. According to the company, "Users can interact and annotate the maps using gesture inputs, enabling them to view the terrain from multiple perspectives and locate tactical advantages in the field. Further, the sandbox facilitates remote networking capabilities, enabling